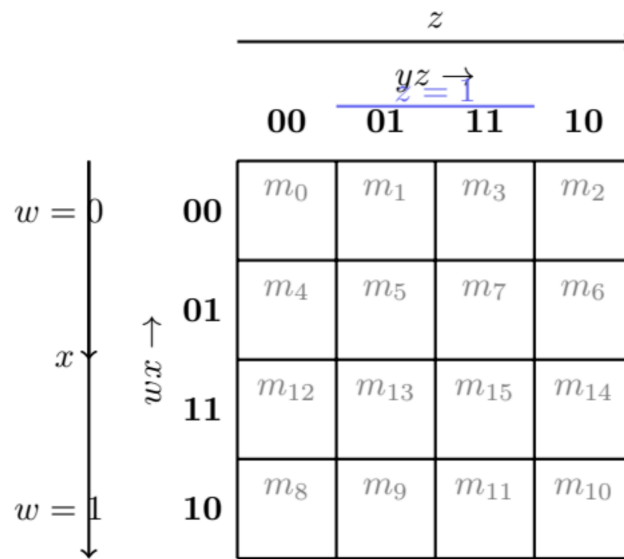


Implicant: any valid group of 1s on the K-map. Any single 1, or any power-of-2 rectangle of 1s. Every group you could ever circle is an implicant.

Prime implicant (PI): any implicant that cannot be made any larger. You cannot add more adjacent 1s to it without pulling in a 0. In other words, it is already as big as it can possibly be.

Essential prime implicant (EPI): a prime implicant that is the only PI covering at least one particular minterm. It is a group whose 1s are unique and not in any other group.

4-Variable K-Map:



Similar to the 3-var KM, the 4-var KM uses a specific input order on the rows and columns to ensure that every pair of physically adjacent cells truly differs by only one variable. If we used normal binary order (00, 01, 10, 11) some neighboring cells would differ in two bits, breaking the whole simplification idea.

4-var KM adjacency rules:

1. **Left $\leftrightarrow$ Right:** m_0 is adjacent to m_2 ; m_4 is adjacent to m_6 ; m_{12} is adjacent to m_{14} ; m_8 is adjacent to m_{10} .
2. **Top $\leftrightarrow$ Bottom:** m_0 is adjacent to m_8 ; m_1 is adjacent to m_9 ; etc.
3. **The four corners** (m_0, m_2, m_8, m_{10}) form a valid group of 4. They are all mutually adjacent when you account for wrapping.

Because we have a lot more minterm squares for 4-var KM, we can have groups as big as 8 or even 16 minterms. Each group represents how many terms we will have in the simplified expression. Always make groups as large as possible. Bigger groups eliminate more variables and give simpler terms. It is always legal to reuse a cell in more than one group.

Simplify $F(w, x, y, z) = \sum(0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14)$.

Step 1: Fill in the map. Place a 1 in each cell corresponding to the listed minterms.

		yz				
		00	01	11	10	
wx	00	1	1	0	1	$\bar{w}\bar{z}$
	01	1	1	0	1	
	11	1	1	0	1	$x\bar{z}$
	10	1	1	0	1	

$\bar{y}$ (8 cells)

Step 2: Find groups.

Group 1 (blue) — 8 cells, columns 00 and 01 (all four rows):

$m_0, m_1, m_4, m_5, m_8, m_9, m_{12}, m_{13}$.

Check each variable: w varies (0 and 1) $\rightarrow$ cancels. x varies $\rightarrow$ cancels. $y = 0$ always $\rightarrow$ keep as $\bar{y}$. z varies $\rightarrow$ cancels. **Term:** $\bar{y}$

Group 2 (red, dashed) — 4 cells using left+right wrap: m_0, m_2, m_4, m_6 (columns 00 and 10, rows 00 and 01).

$w = 0$ always $\rightarrow \bar{w}$. x varies $\rightarrow$ cancels. y varies $\rightarrow$ cancels. $z = 0$ always $\rightarrow \bar{z}$. **Term:** $\bar{w}\bar{z}$

Group 3 (orange, dotted) — 4 cells using left+right wrap: m_4, m_6, m_{12}, m_{14} (columns 00 and 10, rows 01 and 11).

w varies $\rightarrow$ cancels. $x = 1$ always $\rightarrow x$. y varies $\rightarrow$ cancels. $z = 0$ always $\rightarrow \bar{z}$. **Term:** $x\bar{z}$

Step 3: OR the terms together.

$$F = \bar{y} + \bar{w}\bar{z} + x\bar{z}$$

Ex)

Given the truth table below, find the minimal sum-of-products expression for $F(A, B, C, D)$.

A	B	C	D	F	A	B	C	D	F
0	0	0	0	0	1	0	0	0	1
0	0	0	1	0	1	0	0	1	1
0	0	1	0	0	1	0	1	0	0
0	0	1	1	0	1	0	1	1	0
0	1	0	0	1	1	1	0	0	1
0	1	0	1	1	1	1	0	1	1
0	1	1	0	0	1	1	1	0	0
0	1	1	1	0	1	1	1	1	0

Step 1: Read off the minterms. Scan the F column for every row where $F = 1$ and note the row number (the decimal value of $ABCD$):

$$F = 1 \text{ at rows: } 4, 5, 8, 9, 12, 13 \implies F(A, B, C, D) = \sum(4, 5, 8, 9, 12, 13)$$

Step 2: Fill the map. Place a 1 in each of those squares and a 0 everywhere else.

		CD				
		00	01	11	10	
AB	00	0	0	0	0	
	01	1	1	0	0	$B\bar{C}$
	11	1	1	0	0	
	10	1	1	0	0	$A\bar{C}$

Step 3: Identify the groups.

Group 1 (blue) — m_4, m_5, m_{12}, m_{13} (rows 01 and 11, columns 00 and 01):

A varies $\rightarrow$ cancels. $B = 1$ always $\rightarrow B$. $C = 0$ always $\rightarrow \bar{C}$. D varies $\rightarrow$ cancels. **Term:** $B\bar{C}$

Group 2 (red, dashed) — m_8, m_9, m_{12}, m_{13} (rows 10 and 11, columns 00 and 01):

$A = 1$ always $\rightarrow A$. B varies $\rightarrow$ cancels. $C = 0$ always $\rightarrow \bar{C}$. D varies $\rightarrow$ cancels. **Term:** $A\bar{C}$
Note that m_{12} and m_{13} are reused in both groups — that is perfectly fine and gives us bigger, simpler groups than if we had split them up.

Step 4: Final answer.

$$F = B\bar{C} + A\bar{C}$$

Ex)

Consider $F(A, B, C, D) = \sum(0, 2, 3, 5, 7, 8, 9, 10, 11, 13, 15)$.

		CD			
		00	01	11	10
AB	00	1	0	1	1
	01	0	1	1	0
	11	0	1	1	0
	10	1	1	1	1

$\bar{B}\bar{D}$ (essential: covers m_0)

BD (ess.)

Essential prime implicants:

- $\bar{B}\bar{D}$: the 4-corner group (m_0, m_2, m_8, m_{10}). m_0 can only be in this group → **essential**.
- BD : the center 2×2 (m_5, m_7, m_{13}, m_{15}). m_5 can only be in this group → **essential**.

These two PIs cover 8 of the 11 minterms. The remaining uncovered 1s are m_3, m_9, m_{11} .

Non-essential choices (multiple valid options for covering the rest):

Option	Additional terms
A	$CD + AD$
B	$CD + AB$
C	$BC + AD$
D	$BC + AB$

All four options are equally valid minimal expressions. Any one gives F with 4 total product terms of 2 literals each.

Ex)